

Jack Basumata

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Skills

Languages: C, C++, Java, Python, Go, SQL, PHP, Dart

Web: HTML, CSS, JS, React, Angular, Node.js, Express, REST APIs, JWT, Express.js, MERN, CRUDE

Databases: MongoDB, MySQL, PostgreSQL

Data Science & AI: Machine Learning, NLP, Generative AI (LLMs, RAG Basics), Data Analysis & Visualization, Model Deployment(Flask)

Core CS: DSA, OOP, DBMS, CN, Compiler Design, Cybersecurity

Tools: Git, GitHub, VS Code, Taiga, AWS, Docker, Postman, Figma, Vite, Vercel

Experience

Full Stack Development Intern | ZeTheta

April 2026 – May 2026

- Developing full-stack web applications using React, Node.js, Express.js, and MongoDB.
- Collaborating on production-grade features following modern software development practices.

AI Engineering Intern | InAmigos

May 2026 – Present

- Working with machine learning frameworks, data preprocessing, model evaluation, and deployment workflows.
- Developing end-to-end AI pipelines involving data preprocessing, semantic retrieval, model evaluation, and inference optimization.

Projects/Works

Engineered a **MERN**-based online proctoring platform with **JWT** authentication, **CryptoJS** encryption, session management, **WebSocket**-driven real-time monitoring, activity tracking, and secure exam lifecycle management. [Link](#)

Orchestrated **YOLOv8-Pose** for kinematic action recognition and **ResNet-18** for zero-shot ReID. Implemented vector cross-product tripwires to log persistent metrics to a stateful **SQLite** backend. [Link](#)

Developed a full-stack MERN chat application leveraging **Socket.IO** for low-latency real-time messaging, **JWT**-based authentication, **RESTful** APIs, **MongoDB** persistence, and scalable client-server communication. [Link](#)

Developed a decoupled system with a **FastAPI** backend for **MPS**-accelerated inference and a **Streamlit** frontend; utilized transformer-based cross-modal embeddings to correlate textual, layout, and visual features. [Link](#)

Built a retrieval-augmented **LLM** application combining transformer embeddings, **Pinecone** vector indexing, semantic search, prompt orchestration, context retrieval, and API-driven inference pipelines.

Developed an end-to-end **NLP** pipeline employing **TF-IDF** vectorization, feature engineering, hyperparameter-tuned **Scikit-learn** classifiers, and statistical text preprocessing for sentiment categorization.

Education

B.Tech in Computer Science and Engineering | Indian Institute of Information Technology Kottayam

August 2023 – Expected April 2027

Activities

Art • Music • Journalism • Content Creation • Vocal Practicioning • Sports • Travel